

Using Cross-Section Data

NE 630

Lesson 05

Prepare: Read FNRP Sections 2.3–2.4. Identify the three reactor-neutron energy ranges and distinguish potential scattering from compound-nucleus reactions before class.

Macro objective

For a neutron of energy E , determine the interactions it can undergo and compute the probability that it undergoes a particular reaction.

Vocabulary: thermal, epithermal, and fast neutron; evaluated data; elastic and inelastic scattering; capture; absorption; fission; total cross section; potential scattering; compound nucleus; resonance; reaction-rate density; conditional probability

1 What do neutrons do?

Interaction bookkeeping

Any interaction removes a neutron from its *uncollided beam*; only some interactions remove the incident neutron from the system. For mutually exclusive reaction channels x ,

$$\sigma_i^j(E) = \sum_x \sigma_x^j(E), \quad \Sigma_x^i(E) = n_i \sigma_x^i(E).$$

Useful grouped cross sections include

$$\sigma_s \approx \sigma_e + \sigma_{in} + \dots, \quad \sigma_a = \sigma_\gamma + \sigma_f + \sigma_\alpha + \dots$$

Common neutron cross sections

σ_x	reaction or interaction channel
σ_n or σ_e	elastic scattering
$\sigma_{n'}$ or σ_{in}	inelastic scattering
σ_{2n}	$(n, 2n)$ and related multiplicative reactions
σ_γ	radiative capture, (n, γ)
σ_f	fission
σ_α	(n, α)
σ_a	absorption: $\sigma_\gamma + \sigma_f + \sigma_\alpha + \dots$
σ_t	total: sum over all exclusive channels

Beam loss versus system loss

Elastic scattering removes a neutron from the original beam: It destroys that neutron: Which listed reactions can increase the neutron population?

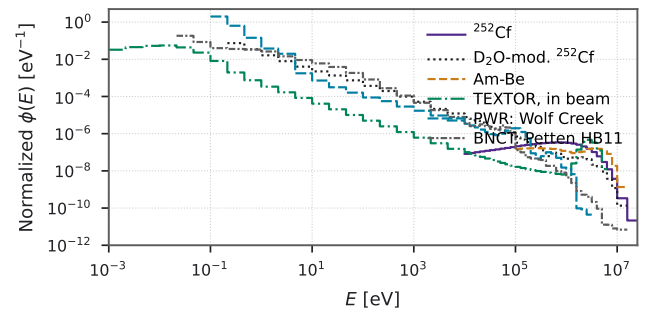
2 Neutrons span many energies

Reactor-neutron energy ranges

range	interval	physical landmark
thermal	$10^{-3} \text{ eV} \leq E < 1 \text{ eV}$	target thermal motion matters
epithermal	$1 \text{ eV} \leq E < 0.1 \text{ MeV}$	many resolved and unresolved resonances
fast	$0.1 \text{ MeV} \leq E \leq 10 \text{ MeV}$	fission-birth-energy range

The useful range spans about ten orders of magnitude. At room temperature, $kT \approx 0.0253 \text{ eV}$; fission neutrons are born with mean energy about 2 MeV.

Examples of normalized neutron spectra



Do not confuse the two functions

A spectrum $\phi(E)$ describes at energy E ; a cross section $\sigma_x^i(E)$ describes for nuclide i and channel x .

3 From evaluated data to a rate

Use one consistent neutron energy

For nuclide i and reaction x , follow

$$\sigma_x^i(E) \xrightarrow{\times n_i} \Sigma_x^i(E) \xrightarrow{\times \phi} R_x^i(E)$$

$$[\sigma] = \text{cm}^2, \quad [\Sigma] = \text{cm}^{-1}, \quad [R] = \text{reactions cm}^{-3} \text{ s}^{-1}.$$

Before summing, verify that every σ_x^i is evaluated at the same E and that the reaction categories do not overlap.

4 The balancing act

Decompose the total reaction-rate density

For a monoenergetic flux ϕ ,

$$R_t = \sum_i \phi \left(\sum_i \sigma_t^i \right) \phi = \left(\sum_i n_i \sigma_t^i \right) \phi$$

$$= \left(\sum_i n_i \sum_x \sigma_x^i \right) \phi = \sum_i \sum_x R_x^i.$$

The useful partial sums are

$$R_x = \sum_i R_x^i, \quad R_t^i = \sum_x R_x^i, \quad R_t = \sum_x R_x = \sum_i R_t^i.$$

Read each rate ratio as a conditional probability

$$\frac{R_x}{R_t} : \text{reaction is } x, \text{ given an interaction,}$$

$$\frac{R_t^i}{R_t} : \text{target is } i, \text{ given an interaction,}$$

$$\frac{R_x^i}{R_t^i} : \text{reaction is } \dots\dots\dots, \text{ given target } i,$$

$$\frac{R_x^i}{R_x} : \text{target is } \dots\dots\dots, \text{ given reaction } x,$$

$$\frac{R_f}{R_a} : \text{an absorption produces } \dots\dots\dots$$

The common flux cancels from these ratios. For a homogeneous mixture at a fixed E ,

$$P(x | \text{interaction}) = \frac{\sum_i n_i \sigma_x^i}{\sum_i n_i \sigma_t^i}.$$

Fission-neutron production

$$\dot{n}_{\text{birth}} = \sum_i \bar{\nu}^i R_f^i, \quad \bar{\nu}_{\text{mix}} = \frac{\sum_i \bar{\nu}^i R_f^i}{R_f}.$$

The first expression is a neutron-production-rate density; the second is the rate-weighted mean neutrons born per fission.

Conditional is not absolute

Over a uniform path of length L , combine Lesson 4 with the rate ratio:

$$P(\text{first interaction is } x \text{ by } L) = \dots\dots\dots$$

5 Apply the bookkeeping

A one-to-one $\text{H}_2\text{O}-^{235}\text{U}$ mixture

At the selected neutron energy, take one water molecule per ^{235}U atom. Thus $n_{\text{H}} : n_{\text{O}} : n_{\text{U}} = 2 : 1 : 1$; write the three as $2n_0$, n_0 , and n_0 . Unlisted cross sections are zero in this exercise.

nuclide	σ_e	σ_f	σ_a
H-1	20 b	0.1 b	—
O-16	4 b	—	—
^{235}U	15 b	100 b	500 b

Build the competing rate strengths

Because every term contains $n_0\phi_0$, compare the effective microscopic strengths:

$$\frac{R_e}{n_0\phi_0} = 2(20) + 1(4) + 1(15) = \dots\dots\dots \text{ b,}$$

$$\frac{R_a}{n_0\phi_0} = 2(0.1) + 1(100 + 500) = \dots\dots\dots \text{ b,}$$

$$\frac{R_f}{n_0\phi_0} = \dots\dots\dots + \dots\dots\dots = \dots\dots\dots \text{ b.}$$

If a neutron interacts, what happens?

$$P(e | \text{int}) = \frac{59}{659.2} = \dots\dots\dots \approx \dots\dots\dots,$$

$$P(\text{H elastic} | \text{int}) = \frac{2(20)}{659.2} = \dots\dots\dots \approx \dots\dots\dots,$$

$$P(\text{U absorption} | \text{int}) = \frac{100 + 500}{659.2} = \dots\dots\dots \approx \dots\dots\dots,$$

$$P(f | \text{absorbed}) = \frac{500}{600.2} = \dots\dots\dots \approx \dots\dots\dots$$

The reusable recipe

At a fixed energy: read $\sigma_x^i(E)$, weight by each $\dots\dots\dots$, form reaction rates, then normalize the desired partial rate by the rate named in the condition.